Xinqi Huang

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Research Interests

Extremal graph theory; extremal set theory; coding theory; additive combinatorics.

Education

Ph.D. in Mathematics September 2022–present

University of Science and Technology of China, Hefei, China

Expected completion: June 2027

Advisors: Prof. Hong Liu, Prof. Tuan Tran, and Prof. Xiande Zhang

B.Sc. in Mathematics September 2018–July 2022

University of Science and Technology of China, Hefei, China

Research Experience

Visiting Student Researcher July–September 2024; May 2025–present

Extremal Combinatorics and Probability Group (ECOPRO)

Institute for Basic Science, Daejeon, South Korea

Publications

1. **Xinqi Huang** and Hong Liu. *Minimum-degree thresholds beyond bounded colourability*. Survey article, accepted for publication in the *Proceedings of the International Consortium of Chinese Mathematicians (ICCM 2025)*. arXiv:2610.39759, 2026.
2. **Xinqi Huang**, Hong Liu, Mingyuan Rong, and Zixiang Xu. *Interpolating chromatic and homomorphism thresholds*. Journal of Combinatorial Theory, Series B, 182:286–324, 2027. doi:10.1016/j.jctb.2026.10.002.
3. **Xinqi Huang**, Chong Shangguan, Xiande Zhang, and Yuhao Zhao. *Focal-free uniform hypergraphs and codes*. Journal of Combinatorial Theory, Series A, 223:106193, 2026. doi:10.1016/j.jcta.2026.106193.
4. Xinbu Cheng, **Xinqi Huang**, Mingyuan Rong, and Zixiang Xu. *Sublinear hitting sets for some geometric graphs*. Frontiers in Combinatorics and Number Theory, 2:23–42, 2026. doi:10.3934/fcnt.2026010.

Preprints

1. **Xinqi Huang** and Hong Liu. *The asymptotic Erdős–Frankl–Füredi cover-free conjecture*. arXiv:2610.22021, 2026.
2. Ting-Wei Chao, **Xinqi Huang**, and Hong Liu. *Asymptotics of the Brown–Erdős–Sós problem at integer exponents*. arXiv:2609.38115, 2026.
3. Lior Gishboliner, **Xinqi Huang**, and Hong Liu. *Homomorphism and VC-dimension thresholds: spectra and separations*. arXiv:2608.24623, 2026.
4. **Xinqi Huang**, Mingyuan Rong, and Chong Shangguan. *Exact homomorphism thresholds beyond cliques*. arXiv:2607.28241, 2026.
5. **Xinqi Huang**, Hong Liu, and Mingyuan Rong. *On the spectrum and structure of blowup thresholds*. arXiv:2607.03035, 2026.
6. Xinbu Cheng, **Xinqi Huang**, Hong Liu, Bin Wang, and Zhifei Yan. *Colour diversity in spanning structures under Dirac-type conditions*. arXiv:2602.23801, 2026.

7. Wenjie Zhong, **Xinqi Huang**, and Xiande Zhang. *Quantitative frameproof codes and hypergraphs*. arXiv:2511.17912, 2025.
8. **Xinqi Huang**, Yuzhen Qi, Mingyuan Rong, and Zixiang Xu. *Largest dyadic dual VC-dimension of non-piercing families*. arXiv:2506.13606, 2025.

Selected Talks

1. *Recent developments in the Brown–Erdős–Sós problem*. Invited talk, DIMACS-FU, Fuzhou University, Fuzhou, China. September 20, 2026.
2. *Recent developments in the Brown–Erdős–Sós problem*. Invited talk, Seminar on Discrete Mathematics, Xiamen University, Xiamen, China. September 14, 2026.
3. *Accumulation points of homomorphism thresholds*. Contributed talk, ASIACOMB 2026, Daejeon, South Korea. August 28, 2026.
4. *On the blowup threshold of graphs*. Invited talk, Young Scholars Forum on Graph Theory and Combinatorics, Shandong University, online. June 6–7, 2026.
5. *Chromatic, homomorphism, blowup thresholds*. Invited talk, Seminars on Discrete Mathematics, Xiamen University, Xiamen, Fujian, China. December 30, 2025.
6. *Chromatic, homomorphism, blowup thresholds and beyond*. Invited talk, Beijing Institute of Technology, Beijing, China. December 15, 2025.
7. *Interpolating chromatic and homomorphism thresholds*. Invited talk, 2025 Anhui SIAM Workshop, Wuhu, Anhui, China. April 18–20, 2025.
8. *Interpolating chromatic and homomorphism thresholds*. Invited talk, SCMS Combinatorics Seminar, Fudan University, Shanghai, China. March 25, 2025.

Awards and Honors

National Scholarship for Graduate Students 2026
Ministry of Education of the People's Republic of China

National Scholarship for Graduate Students 2024
Ministry of Education of the People's Republic of China

Outstanding Graduate 2022
University of Science and Technology of China

Teaching Experience

Course Assistant, Selected Topics in Combinatorics, USTC Spring 2024
Helped select recent research papers, organize course themes, and present several lectures.

Teaching Assistant, Combinatorics, USTC Fall 2021
Led problem sessions for the course.

Teaching Assistant, Linear Algebra, USTC Spring 2020
Led problem sessions for the course.